

CLARUS Analysis Report

Document: rfc2460.txt
Analyzed: 2026-05-06T15:16:44.439081

1. Document Processing

Metric	Value
Total Segments	903
Normative Segments	114
Informative Segments	527
Unknown Segments	262
High Confidence (≥ 0.7)	169
Low Confidence (< 0.4)	0
Text Length (chars)	88077

2. Semantic Extraction

Metric	Value
Total Profiles	10
Average Confidence	70.5%

Profiles (10)

#	Sentence	Subject	Object	Modality	Condition
1	highest-order two bits specify the action that must be taken if the	action that	taken if the	must	the
2	Options area of a header. If more than one octet of padding is required	padding is	used	should	more than one octet of padding is required
3	If Segments Left is non-zero, the node must discard the packet and ...	packet and	n ICMP Parameter Problem	must	Segments Left is non-zero
4	The following error conditions may arise when reassembling fragmented packets	error conditions	may arise when reassembling fragmented	may	reassembling fragmented
5	packet must be discarded. If the first fragment (i.e., the one	one	discarded	must	the first fragment (i
6	o If any other action is desired, the information must be encoded	any other action	encoded	must	any other action is desired
7	lifetime. That is the reason the IPv4 "Time to Live" field is...	in practice	ought to	ought to	any
8	an earlier flow whose lifetime may not have expired yet. This may ...	have expired yet	may not	may not	the minimum time for rebooting the node is known
9	o It may be assumed that, when either of the option-bearing headers...	assumed that	may	may	either of the option-bearing headers are present
10	This document and translations of it may be copied and furnished to...	copied and furnished to others	may	may	the above copyright notice and this paragraph included on all such copies and derivative works

3. Terminology Analysis

Metric	Value
Defined Terms	0
Undefined Terms	50
Definition Coverage	0.0%

4. Contradiction Analysis

Total contradictions detected: **83**

#	Type	Conf.	Statement A	Statement B
1	semantic	100%	Its alignment requirement is $8n+2$, to ensure that th...	Its alignment requirement is $4n+3$, to ensure that th...
2	semantic	100%	highest-order two bits specify the action that must ...	entire Option Data field must be treated as zero-val...
3	semantic	100%	o If any other action is desired, the information m...	contains 59, those octets must be ignored, and passe...
4	semantic	100%	The sequence of options within a header must be proc...	00 - skip over this option and continue processing t...
5	semantic	100%	65,535 octets, then that fragment must be discarded ...	contains 59, those octets must be ignored, and passe...
6	semantic	100%	flow's path must be specified as part of the de...	must be careful not to use a flow label that it migh...
7	semantic	100%	of 8 octets in length. These padding options must b...	not have length and value fields.
8	semantic	100%	ensure that multi-octet values within Option Data fi...	not have length and value fields.
9	semantic	100%	If an option X required two data fields, one of leng...	+++++ Option Type=Y +++++
10	semantic	100%	accept fragmented packets that reassemble to more th...	not send packets larger than 1500 octets unless it h...
11	semantic	100%	addresses from the IP header in its checksum computa...	computation yields a result of zero, it must be chan...
12	semantic	100%	10 - discard the packet and, regardless of whether o...	11 - discard the packet and, only if the packet'...
13	semantic	99%	Its alignment requirement is $8n+2$, to ensure that th...	+++++
14	semantic	99%	Its alignment requirement is $8n+2$, to ensure that th...	+++++
15	semantic	99%	Its alignment requirement is $4n+3$, to ensure that th...	+++++
16	semantic	99%	not required, to verify that these conditions are sa...	If an option X required two data fields, one of leng...
17	semantic	99%	not required, to verify that these conditions are sa...	or to design a header compression scheme that only w...
18	semantic	99%	+++++	+++++
19	semantic	99%	+++++	+++++
20	semantic	99%	+++++	+++++
21	semantic	99%	+++++	+++++
22	semantic	99%	fragment of that packet, reassembly of that packet m...	of that fragment is 1, then that fragment must be di...
23	semantic	99%	+++++	header with an unrecognized Routing Type value, the ...
24	semantic	99%	adjust its packets to fit the measured path MTU (i.e...	is not required to reduce the size of subsequent pac...
25	semantic	98%	a Routing header must send an ICMP Packet Too Big me...	from the set of fields that must remain the same for...

26	semantic	98%	o If any other action is desired, the information m...	packet in one piece, link-specific fragmentation and...
27	semantic	98%	00 - skip over this option and continue processing t...	ensure that multi-octet values within Option Data fi...
28	semantic	98%	constraints relative to the above listed headers mus...	IPv6 nodes must accept and attempt to process extens...
29	semantic	98%	packets includes a Hop-by-Hop Options header, then t...	must be careful not to use a flow label that it migh...
30	semantic	98%	originating a UDP packet, an IPv6 node must compute ...	Unlike IPv4, IPv6 nodes are not required to enforce ...
31	semantic	98%	+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+...	destination. At the originating node, that address ...
32	semantic	98%	packets includes a Hop-by-Hop Options header, then t...	If any of those packets includes a Routing header, t...
33	semantic	97%	+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+...	+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+...
34	semantic	97%	+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+...	+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+...
35	semantic	97%	+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+...	+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+...
36	semantic	97%	+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+...	+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+...
37	semantic	97%	+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+...	+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+...
38	semantic	95%	appear immediately after an IPv6 header only. Nonet...	00 - skip over this option and continue processing t...
39	semantic	93%	All packets belonging to the same flow must be sent ...	must be careful not to use a flow label that it migh...
40	semantic	93%	o The service interface to the IPv6 service within ...	layer protocol. The default value must be zero for ...
41	semantic	90%	(optional) data confidentiality are specified for IPv6.	The IPv6 version of ICMP, which all IPv6 implementat...
42	modality	90%	constraints relative to the above listed headers mus...	The sequence of options within a header must be proc...
43	modality	90%	IPv6 nodes must accept and attempt to process extens...	The sequence of options within a header must be proc...
44	modality	90%	The sequence of options within a header must be proc...	highest-order two bits specify the action that must ...
45	modality	90%	The sequence of options within a header must be proc...	entire Option Data field must be treated as zero-val...
46	modality	90%	If Segments Left is zero, the node must ignore the R...	Multicast addresses must not appear in a Routing hea...
47	modality	90%	If Segments Left is non-zero, the node must discard ...	Multicast addresses must not appear in a Routing hea...
48	modality	90%	must discard the packet and send an ICMP Packet Too...	Multicast addresses must not appear in a Routing hea...
49	modality	90%	contains 59, those octets must be ignored, and passe...	1661)) must be configured to have an MTU of at least...
50	modality	90%	IPv6 requires that every link in the internet have a...	1661)) must be configured to have an MTU of at least...
51	modality	90%	IPv6 requires that every link in the internet have a...	accept fragmented packets that reassemble to more th...
52	modality	90%	1661)) must be configured to have an MTU of at least...	adjust its packets to fit the measured path MTU (i.e...
53	modality	90%	1661)) must be configured to have an MTU of at least...	accept fragmented packets that reassemble to more th...
54	modality	90%	1661)) must be configured to have an MTU of at least...	not send packets larger than 1500 octets unless it h...
55	modality	90%	adjust its packets to fit the measured path MTU (i.e...	accept fragmented packets that reassemble to more th...
56	modality	90%	accept fragmented packets that reassemble to more th...	1280, but must include a Fragment header in those pa...
57	modality	90%	is not required to reduce the size of subsequent pac...	required to set the field to zero when originating a...
58	modality	90%	o The service interface to the IPv6 service within ...	o An upper-layer protocol must not assume that the ...
59	modality	90%	layer protocol. The default value must be zero for ...	o An upper-layer protocol must not assume that the ...
60	modality	90%	o An upper-layer protocol must not assume that the ...	addresses from the IP header in its checksum computa...
61	modality	90%	originating a UDP packet, an IPv6 node must compute ...	packet(s) must not include a Routing header that was...
62	modality	90%	computation yields a result of zero, it must be chan...	packet(s) must not include a Routing header that was...

63	modality	90%	FFFF for placement in the UDP header. IPv6 receiver...	packet(s) must not include a Routing header that was...
64	modality	90%	data, an upper-layer protocol must take into account...	packet(s) must not include a Routing header that was...
65	modality	90%	MSS must be computed as the maximum packet size min...	packet(s) must not include a Routing header that was...
66	modality	90%	packet(s) must not include a Routing header that was...	flow labels must be chosen (pseudo-)randomly and uni...
67	modality	90%	packet(s) must not include a Routing header that was...	All packets belonging to the same flow must be sent ...
68	modality	90%	packet(s) must not include a Routing header that was...	packets includes a Hop-by-Hop Options header, then t...
69	modality	90%	packet(s) must not include a Routing header that was...	If any of those packets includes a Routing header, t...
70	modality	90%	flow labels must be chosen (pseudo-)randomly and uni...	protocol or the flow-setup hop-by-hop option. A sou...
71	modality	90%	All packets belonging to the same flow must be sent ...	protocol or the flow-setup hop-by-hop option. A sou...
72	modality	90%	packets includes a Hop-by-Hop Options header, then t...	protocol or the flow-setup hop-by-hop option. A sou...
73	modality	90%	If any of those packets includes a Routing header, t...	protocol or the flow-setup hop-by-hop option. A sou...
74	modality	90%	flow's path must be specified as part of the de...	protocol or the flow-setup hop-by-hop option. A sou...
75	modality	90%	protocol or the flow-setup hop-by-hop option. A sou...	must be careful not to use a flow label that it migh...
76	modality	90%	a Routing header must send an ICMP Packet Too Big m...	must not include the reverse of the Routing header i...
77	modality	90%	from the set of fields that must remain the same for...	must not include the reverse of the Routing header i...
78	semantic	87%	If Segments Left is zero, the node must ignore the R...	must discard the packet and send an ICMP Packet Too ...
79	semantic	82%	+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+...	+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+...
80	semantic	82%	+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+...	+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+...
81	semantic	82%	send an ICMP Parameter Problem, Code 0, message to	send an ICMP Time Exceeded -- Hop Limit Exceeded in
82	semantic	82%	send an ICMP Parameter Problem, Code 0, message to	send an ICMP Time Exceeded -- Hop Limit Exceeded in
83	semantic	80%	This document specifies the basic IPv6 header and th...	semantics of IPv6 addresses are specified separately...